

SCHOOL OF COMPUTER & SYSTEMS SCIENCES, JNU
MCA I Semester Final Examination 2022-23
Discrete Mathematics

Time: 2 Hrs.

Total Marks: 60

Note: Attempt ALL Questions. Do Not Write Anything on the Question Paper.

Q. 1 If R is a relation "Less Than" from $A = \{1, 2, 3, 4\}$ to $B = \{1, 3, 5\}$ then what will be $R \circ R^{-1}$ (6)

Q. 2 R is a relation on set N defined by $\{(x, y) : x, y \in N, x + 2y = 21\}$. Determine the type of relation R ? (6)

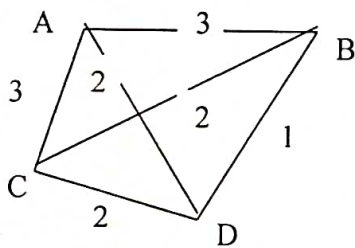
Q. 3 Define Boolean algebra. Prove that the power set of any set forms a Boolean algebra. (6)

Q. 4 There is a graph of 25 vertex (order) and 62 edges (size). The degree of every vertex is 3, 4, 5 or 6. There are 2 vertices of degree 4 and 11 vertices of degree 6. How many vertices of graph have degree 5? (6)

Q. 5 What is a spanning tree of a graph? How to obtain fundamental circuits, given a spanning tree? How many fundamental circuits exist in a graph? (6)

Q. 6 Explain the ringsum, fusion and deletion operations on graphs giving suitable examples. (6)

Q. 7 For the given graph, find the minimal spanning tree using Prim's algorithm (with steps). (6)



Q. 8 What do you understand by the planarity of a graph? Explain with example. State and prove the Euler formula to check the planarity of a graph. (6)

Q. 9 Define a Deterministic Finite Automaton (DFA)? Design a DFA for sentences in $\{a, b\}$ such that every a has b immediately to its right. (6)

Q. 10 Determine the language accepted by the following automaton. (6)

